Phishing Email Detector using Machine Learning

ENSIA – CNS Module | Team 18

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Problem

- **Phishing emails are increasing** rapidly
- Users get tricked into:
  - Giving away passwords
  - Clicking malicious links
- Traditional filters are not enough

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Phishing attacks
are among the most
common cyber threats

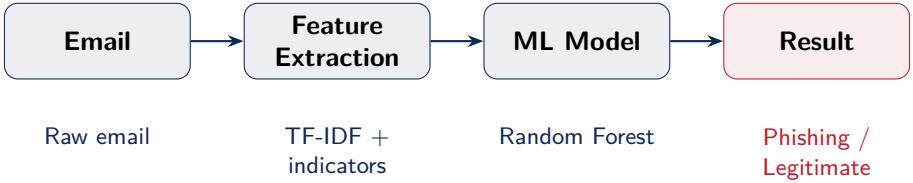
Our Solution

- **Intelligent phishing detector**
- Uses **Machine Learning**
- Integrated as a **browser extension**
- Works **directly in Gmail**

Key Idea

Automated, real-time analysis of incoming emails — no manual intervention needed.

How It Works



1. Extract text from email
2. Generate features (TF-IDF + indicators)
3. Predict: **Phishing** or **Legitimate**

Features Used

- Suspicious URLs
- Urgent words *(urgent, verify, click)*
- Number of links
- Exclamation marks
- TF-IDF text representation

Feature Engineering

A combination of **linguistic cues** and **structural indicators** gives the model rich signal for classification.

Models

Model	Evaluated	Selected
Logistic Regression	✓	
Random Forest	✓	✓

Why Random Forest?

Offers better generalization, handles non-linear patterns, and is more robust to noisy features compared to Logistic Regression.

Results

98.5%
Accuracy

0.979
F1-Score

- **Only 12 errors** out of 777 emails
- Excellent precision and recall balance

System Architecture



Browser Extension

Reads email content from Gmail and displays results.

Flask Backend

Receives requests, runs preprocessing, and queries the model.

ML Model

Trained Random Forest that returns the classification.

Final Result — User View

✓ Legitimate

Confidence: **97%**

No suspicious indicators found

× Phishing

Confidence: **99%**

Suspicious URL, urgent language

- **Green** → Legitimate
- **Red** → Phishing
- Shows **confidence score**
- Shows **reasons** for the classification

Limitations

- Only **English emails** supported
- **Text-only** analysis — images and attachments not inspected
- Requires a **local server** to run

Future Work

Multi-language support, image/attachment analysis, and cloud deployment.

Conclusion

- **Effective phishing detection** with 98.5% accuracy
- **Real-time integration** directly inside Gmail
- **Easy to use** — transparent to the end user

Protecting users from phishing — one email at a time.

Thank You

Questions?

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